

Direct Memory Access (DMA)

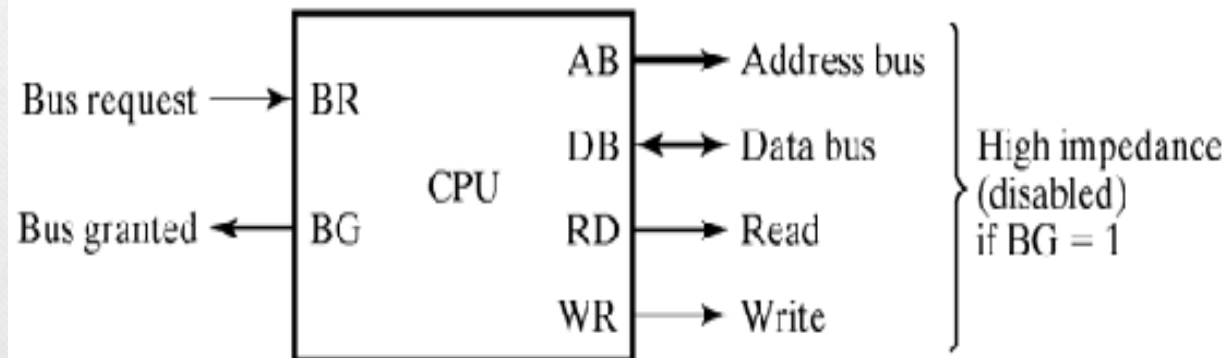
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DMA (DIRECT MEMORY ACCESS)

NEED

- Peripheral devices manage the memory bus directly and improve the speed of transfer between peripheral device and memory without the intervention of CPU.

CPU Bus signals for DMA



Two control signals used to facilitate the DMA transfer:

- BUS REQUEST
- BUS GRANT

BUS REQUEST:

- Signal from DMA controller to the CPU requesting to relinquish control of the buses.
- When this input is high CPU stops executing the instruction and places the address, data, read and write lines in high impedance.

BUS GRANT:

- Signal from CPU to DMA controller to the buses are in high impedance state.
- DMA takes the control of buses and memory transfer occur without the intervention of CPU.

At the end of the transfer, bus request line is disabled followed by bus grant signal disable and CPU returns to its normal operation

DMA TRANSFER MODES

- BURST TRANSFER
- CYCLE STEALING

BURST TRANSFER:

- block consisting of number of words is transferred in a continuous burst while

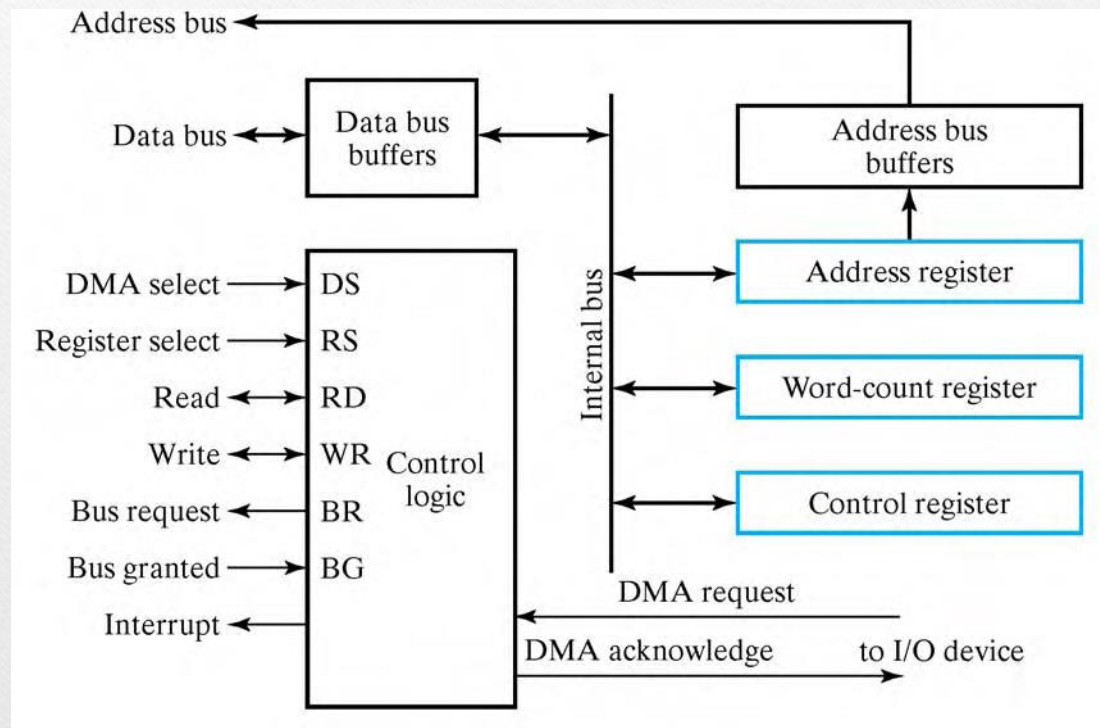
DMA controller is master of memory buses.

- used in high speed peripheral devices such as magnetic disk.

CYCLE STEALING:

- transfers one data word at a time after which it must return the control of the buses to CPU.
- CPU delays its operation for one memory cycle to allow the direct memory transfer to steal one memory cycle

DMA CONTROLLER



-DMA controller is similar to an interface and is used to communicate with the CPU

and IO device.

-Communication to the CPU is established by the data and control lines.

-Communication with IO device is established by DMA request and Acknowledge.

-when $BG = 0$, CPU communicate with DMA register through data bus to read or write to DMA registers.

-When $BG = 1$, CPU relinquished the buses and the DMA can communicate directly with memory by specifying address in the address bus and enabling read/write control signals.

-DMA controller consists of address, word count and control register.

Address Register: contains address to specify the desired location in memory and is incremented after each word is transferred to memory.

Word Count Register: holds number of words to be transferred and decremented after each transfer and tested for zero.

Control Register: specifies the mode of transfer. f

-DMA is initialized by the CPU by executing a program consisting of IO instructions that include the address for selecting particular DMA registers.

-CPU initializes the DMA by sending the following information.

1. Starting address of memory block where data are available for read and where data to be stored in write operation.

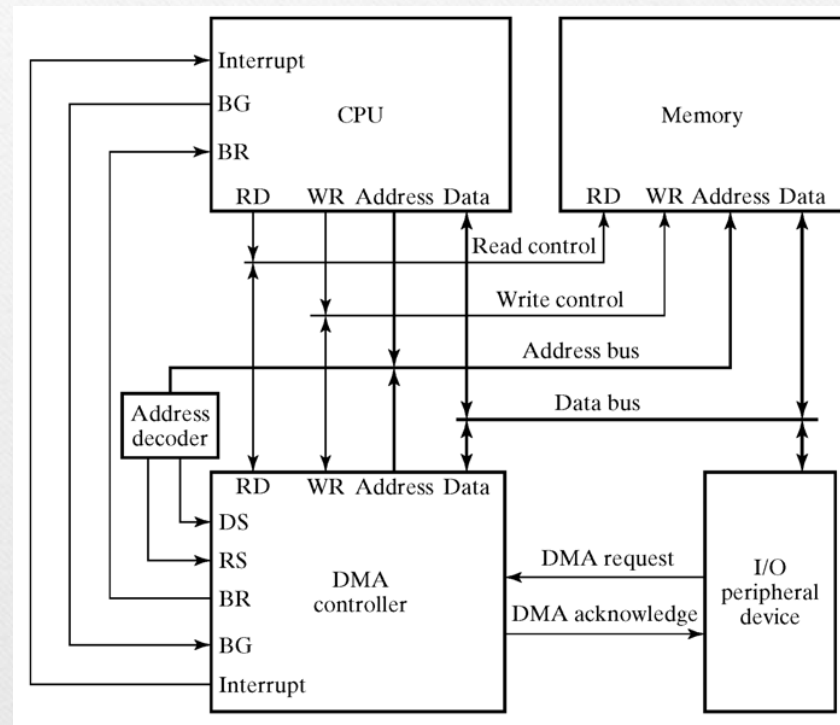
2. The word count – specifies the number of words in memory block

3. Control to specify the mode of transfer such read or write.

4. Control to start the DMA transfer.

- Once the DMA is initialized, CPU stops communicating with the DMA, and DMA now starts and continues to transfer data between memory and peripheral unit until an entire block is transferred.

DMA TRANSFER



- DMA has its own address, which activates the DS and RS lines.
- CPU initializes the DMA through the data bus.
- When peripheral device sends a DMA request, DMA controller activates BR, informing CPU to relinquish the buses. CPU responds by enabling BG.
- DMA transfers the value of address register in to address bus and enables DMA Acknowledge.
- Peripheral devices transfers the word in to data bus (write) or receives a word (read).
- After each word is transferred, DMA increments the address register and decrements its word count register.
- If the word count doesn't reach zero, DMA checks the request line from peripheral devices.
- If the peripheral speed is slow, the DMA request line may come somewhat later. In this case DMA disables the BR line to CPU, so that CPU continues to execute its program.
- Word count reaches zero, DMA stops transfer and removes BR to CPU.

- DMA controller may have more than one channel. Each channel has a request and acknowledge pair of control signals which are connected to separate peripheral devices and has its own address register and word count register.

- Priority is established among the channels, such that high priority channels are serviced first.

Application:

- Fast transfer of information between magnetic disks and memory
- Updating the display in an interactive terminal.

- Consider a computer system with a **CPU**, an **I/O device D**, and **main memory M** connected via a shared bus with a data bus width of one word (16 bits). The CPU can execute a maximum of **1,000,000 instructions per second**. On average, an instruction requires **five processor cycles**, with three of those cycles using the memory bus. A memory read or write operation takes **one processor cycle**. Very large blocks of data need to be transferred between M and D. Calculate the **I/O data transfer rate**, in words per second, possible through D if DMA transfer is used. Assume that the DMA module can use all available cycles and ignore any setup or status-checking time.

- Solution:**

The CPU is busy executing instructions **95%** of the time, leaving **5%** of the time for other tasks.

During the 95% time, DMA can use **0.95×3 cycles** (since memory operations occupy 3 cycles).

During the remaining 5% time, when the CPU isn't executing instructions, all cycles are available for DMA, so it uses **0.05×6 cycles**.

Total cycles available for DMA = **$(0.05 \times 6 + 0.95 \times 3)$ cycles**.

In 1 second, the CPU can execute **1,000,000 instructions**.

Therefore, the rate of transfer = $1,000,000 \times (0.05 \times 6 + 0.95 \times 3) = 3.15 \times 10^6$ words per second¹.

- A hard disk with a transfer rate of **1 Mbyte/second** constantly transfers data to memory using DMA.
- The processor runs at **500 MHz** and takes **500 clock cycles** to initiate and **1000 clock cycles** to complete a DMA transfer.
- If the size of the transfer is **1 Kbyte**, what percentage of processor time is consumed for the transfer operation?
- **Solution:**
- The total time for DMA transfer = **500 + 1000 = 1500 clock cycles**.
- The processor runs at **500 MHz**, so each clock cycle takes **2 ns**.
- Total time for DMA transfer = **1500 × 2 ns = 3000 ns**.
- Transfer size = **1 Kbyte = 1024 bytes**.
- Transfer rate = **1 Mbyte/second = 1024 Kbytes/second**.
- Time taken for transfer = **1024 bytes / 1024 Kbytes/second = 1 ms**.
- Percentage of processor time consumed = (3000 ns / 1 ms) × 100% = 0.3%

- Consider a device operating at **8 Mbps** speed, transferring data to memory using cycle stealing mode of DMA.
- If it takes **250 ns** to transfer **16 bytes** of data to memory when it is ready, what percentage of time is the CPU blocked due to DMA?
- **Solution:**
- Time taken for transfer = **250 ns**.
- Transfer size = **16 bytes**.
- Transfer rate = **8 Mbps** = 8×10^6 bits/second.
- Time taken for transfer = **16 bytes** $\times$ **8 bits/byte** / 8×10^6 bits/second = **20 μ s**.
- Percentage of time CPU is blocked = $(20 \mu\text{s} / 1 \text{ ms}) \times 100\% = 2\%$